

Web Application Development (WAD)

Mock Oral Examination – Questions & Answers

Subject Code: 314458 | T.E. (IT) – 2019 Course

Dept. of Information Technology

Instructions: This document covers oral questions from all 4 assignments. Prepare each question thoroughly as external examiners typically ask from any practical covered in the lab manual.

Assignment 1a – HTML, CSS & Bootstrap Dashboard

Q1. What does HTML stand for?

Ans: HTML stands for HyperText Markup Language. It is the standard markup language used to create the structure of web pages. HTML uses a series of elements/tags to describe the content.

Q2. What is the purpose of `<!DOCTYPE html>`?

Ans: The `<!DOCTYPE html>` declaration tells the browser that the document is an HTML5 document. It must be the very first line in an HTML file before the `<html>` tag.

Q3. What are the main tags used in a basic HTML document?

Ans: The main tags are: `<html>` (root element), `<head>` (metadata), `<title>` (page title shown in browser tab), `<body>` (visible page content), `<h1>`-`<h6>` (headings), and `<p>` (paragraphs).

Q4. What is CSS and why is it used?

Ans: CSS stands for Cascading Style Sheets. It is used to control the visual presentation of HTML elements — including layout, colors, fonts, spacing, and responsiveness. CSS separates content from design, allowing one stylesheet to style multiple pages.

Q5. What is Bootstrap and what problem does it solve?

Ans: Bootstrap is a free, open-source front-end framework that provides ready-made CSS and JavaScript components. It solves cross-browser compatibility issues and makes it easy to create responsive, mobile-first websites without writing all styles from scratch.

Q6. What is a responsive web page?

Ans: A responsive web page automatically adjusts its layout and design to fit different screen sizes — desktop, tablet, or mobile — using techniques like fluid grids, flexible images, and CSS media queries.

Q7. How do you include Bootstrap in your project?

Ans: Bootstrap can be included via CDN links: a CSS link in the `<head>` (`<link rel='stylesheet' href='bootstrap.min.css'>`), the jQuery library script, and the Bootstrap JS script before `</body>`.

Q8. What is a Bootstrap grid system?

Ans: Bootstrap's grid system uses a 12-column layout. You wrap content in a `.container`, divide rows with `.row`, and assign column widths using classes like `.col-md-4` (takes 4 of 12 columns on medium screens).

Q9. What is a card component in Bootstrap?

Ans: A Bootstrap card is a flexible content container with a `.card` class. It can include a header, body, footer, image, and text. Cards are commonly used to display statistics, user profiles, and dashboard widgets.

Q10. What is a sidebar in a web dashboard?

Ans: A sidebar is a vertical navigation panel on the left (or right) side of a web page. It typically contains links/menus for navigating different sections of the application, like in an admin dashboard.

Q11. What are inline vs. block elements in HTML?

Ans: Block elements (like `<div>`, `<p>`, `<h1>`) take up the full width and start on a new line. Inline elements (like `<span>`, `<a>`, `<strong>`) only take up as much width as their content and don't break the line.

Q12. What is the difference between class and id in HTML/CSS?

Ans: An id (`#`) is unique — only one element per page should have a given id. A class (`.`) can be applied to multiple elements. In CSS, `#idname` targets unique elements; `.classname` targets all elements with that class.

Assignment 1b – JavaScript, AJAX & Local Storage

Q1. What is AJAX?

Ans: AJAX stands for Asynchronous JavaScript and XML. It allows web pages to send/receive data from a server asynchronously — without reloading the entire page. This creates faster, more interactive web applications.

Q2. What object is used to make AJAX requests?

Ans: The XMLHttpRequest (XHR) object is used in JavaScript to make AJAX requests. It can send GET or POST requests to a server, and its onreadystatechange event handler processes the server's response.

Q3. What are the readyState values in XMLHttpRequest?

Ans: 0 = Not initialized (open() not called), 1 = Connection established (open() called), 2 = Request received (send() called), 3 = Processing (downloading data), 4 = Complete (response ready).

Q4. What HTTP status code means 'success'?

Ans: HTTP status 200 means 'OK' — the request was successful. Other codes: 403 = Forbidden, 404 = Page Not Found, 500 = Internal Server Error.

Q5. What is the difference between synchronous and asynchronous requests?

Ans: Synchronous requests block the browser until the server responds — the page freezes. Asynchronous requests (used by AJAX) run in the background, allowing the user to continue interacting with the page while data is fetched.

Q6. What is localStorage?

Ans: localStorage is a Web Storage API that lets JavaScript store key-value data persistently in the browser. Data persists even after the browser is closed. It is accessed via localStorage.setItem(), localStorage.getItem(), and localStorage.removeItem().

Q7. How is AJAX different from a regular form POST?

Ans: A regular form POST reloads the entire page with the response. AJAX sends the form data in the background and updates only part of the page dynamically — providing a smoother user experience.

Q8. What methods does XMLHttpRequest provide?

Ans: Key methods: open('method', 'URL', async) — configures the request; send(data) — sends the request; setRequestHeader(x, y) — adds headers; abort() — cancels the request; getAllResponseHeaders() — retrieves response headers.

Q9. What is JSON and how is it used with AJAX?

Ans: JSON (JavaScript Object Notation) is a lightweight data format used to exchange data between client and server. In AJAX, the server often returns JSON data which JavaScript parses with `JSON.parse()` and uses to update the DOM.

Q10. What is DOM manipulation?

Ans: DOM (Document Object Model) manipulation means using JavaScript to dynamically change HTML content and structure. For example, `document.getElementById('id').innerHTML = 'new content'` updates an element's content without reloading the page.

Assignment 2a – Git & GitHub Version Control

Q1. What is Git?

Ans: Git is an open-source distributed version control system created by Linus Torvalds in 2005. It tracks changes in source code, allows multiple developers to collaborate, and lets you revert to previous versions of a project.

Q2. What is the difference between Git and GitHub?

Ans: Git is the version control tool/software that runs locally on your computer. GitHub is a cloud-based hosting platform that stores Git repositories online, enabling collaboration, code sharing, and remote access.

Q3. What is a repository in Git?

Ans: A repository (repo) is a directory where Git tracks all changes to your project files. It contains the entire history of commits. A local repo is on your machine; a remote repo (like on GitHub) is hosted online.

Q4. What is the purpose of 'git init'?

Ans: 'git init' initializes a new empty Git repository in the current folder. It creates a hidden `.git` directory that stores all version control data including commits, branches, and configuration.

Q5. What is the difference between git add and git commit?

Ans: 'git add' moves changed files to the staging area (prepares them for commit). 'git commit' takes a snapshot of all staged files and permanently saves them in the repository history with a message.

Q6. What is a branch in Git?

Ans: A branch is an independent line of development. The default branch is 'main' (or 'master'). You can create branches (git branch name) to work on features without affecting the main code, then merge them back.

Q7. What are git push and git pull?

Ans: 'git push' uploads your local commits to a remote repository (e.g., GitHub). 'git pull' downloads and merges the latest changes from the remote repository into your local branch.

Q8. What is git clone?

Ans: 'git clone <URL>' creates a complete local copy of a remote repository, including all branches and commit history. It is used to download a project from GitHub to your local machine.

Q9. What is git merge?

Ans: 'git merge branch-name' integrates changes from the specified branch into the current branch. It combines the histories of both branches. If the same file was changed in both, a merge conflict may occur.

Q10. What is git status?

Ans: 'git status' shows the current state of the working directory — which files are modified, which are staged for commit, and which are untracked. It helps you understand what changes are pending.

Q11. What is the command to push code to GitHub for the first time?

Ans: First: git remote add origin <repo-URL> (links your local repo to GitHub). Then: git push --set-upstream origin master (pushes the master branch and sets it as the default remote tracking branch).

Assignment 2b – Docker Container Environment

Q1. What is Docker?

Ans: Docker is an open-source platform for developing, shipping, and running applications inside lightweight, portable containers. Containers package an application with all its dependencies so it runs consistently across different environments.

Q2. What is a Docker container?

Ans: A Docker container is a lightweight, standalone executable package that includes everything needed to run an application: code, runtime, libraries, and settings. Containers are isolated but share the host OS kernel.

Q3. What is the difference between a container and a virtual machine?

Ans: A VM virtualizes entire hardware including a full OS (heavy, slow to start). A container virtualizes only at the OS level, sharing the host kernel — making containers much lighter, faster, and more portable.

Q4. What is a Docker image?

Ans: A Docker image is a read-only template used to create containers. It is built from a Dockerfile and contains the application code, dependencies, and configuration. Running an image creates a container.

Q5. What is a Dockerfile?

Ans: A Dockerfile is a text file with instructions to build a Docker image. Common instructions: FROM (base image), RUN (execute commands), COPY (add files), EXPOSE (declare port), CMD (default command to run).

Q6. What command runs a Docker container?

Ans: 'docker run <image-name>' creates and starts a container from an image. Useful flags: -d (detached/background), -p host:container (port mapping), --name (assign a name).

Q7. What is Docker Hub?

Ans: Docker Hub is a public registry/repository for Docker images. Developers can push their images to Docker Hub and pull official or community images (like nginx, node, mongo) using 'docker pull image-name'.

Assignment 2c – Angular Application (Register/Login/Profile)

Q1. What is Angular?

Ans: Angular is a TypeScript-based open-source front-end web framework developed by Google. It is used to build dynamic single-page applications (SPAs) with features like two-way data binding, dependency injection, and a component-based architecture.

Q2. What is a component in Angular?

Ans: A component is the fundamental building block of an Angular application. Each component has a TypeScript class (.ts), an HTML template (.html), and a CSS stylesheet. The @Component decorator defines the selector, template, and style.

Q3. What is the Angular CLI?

Ans: Angular CLI (Command Line Interface) is a tool to create, build, and manage Angular projects. Key commands: 'ng new project-name' (create project), 'ng serve -o' (run and open in browser), 'ng generate component name' (create component).

Q4. What is two-way data binding in Angular?

Ans: Two-way data binding (using [(ngModel)]) synchronizes data between the component's TypeScript class and the HTML template automatically. Changes in the UI update the model, and changes in the model update the UI.

Q5. What is the AppModule in Angular?

Ans: AppModule (app.module.ts) is the root module of an Angular app. It declares all components, imports required Angular modules (like BrowserModule, FormsModule), and bootstraps the root component (AppComponent).

Q6. Why is FormsModule needed in Angular?

Ans: FormsModule must be imported in AppModule to use template-driven forms with directives like ngModel. Without it, Angular cannot recognize ngModel for two-way data binding in forms.

Q7. What is routing in Angular?

Ans: Angular routing enables navigation between different views/components without reloading the page. The RouterModule and Routes are configured in app-routing.module.ts with path-to-component mappings.

Q8. What is the difference between ngModel and ngForm?

Ans: ngModel is used for two-way data binding on individual form controls. ngForm represents the entire form and provides access to form validity state and all control values collectively.

Q9. How do you run an Angular application?

Ans: Use 'ng serve -o' from the Angular CLI in the project directory. This compiles the app and opens it at http://localhost:4200 in the browser. Any code changes are auto-reloaded.

Q10. What is a service in Angular?

Ans: A service is a TypeScript class decorated with `@Injectable` that provides shared logic (like HTTP calls, authentication, data storage) that can be injected into multiple components using dependency injection.

Assignment 3a – Node.js Static Website Server

Q1. What is Node.js?

Ans: Node.js is an open-source, cross-platform JavaScript runtime environment built on Chrome's V8 engine. It allows JavaScript to run on the server side, enabling the development of fast, scalable network and web applications.

Q2. What is Express.js?

Ans: Express.js is a minimal and flexible Node.js web application framework. It simplifies routing, middleware integration, and serving static/dynamic content. It is the most popular framework for building web servers and REST APIs with Node.js.

Q3. How do you serve static files in Express?

Ans: Use `app.use(express.static('folder-name'))` in your Express app. This tells Express to serve all files (HTML, CSS, images, JS) inside the specified folder as static content.

Q4. What is npm and what is it used for?

Ans: npm (Node Package Manager) is the default package manager for Node.js. It is used to install, manage, and share JavaScript packages/libraries. 'npm install package-name' installs a package; 'npm init' creates a package.json.

Q5. What is package.json?

Ans: package.json is a configuration file for Node.js projects. It stores the project name, version, entry point (main), scripts (like 'start': 'node index.js'), and list of dependencies. Created with 'npm init'.

Q6. What does app.listen() do in Express?

Ans: `app.listen(PORT, callback)` starts the Express web server and listens for incoming HTTP requests on the specified port number. The callback runs when the server successfully starts (e.g., `console.log('Server running on port 3000')`).

Q7. What is the difference between GET and POST?

Ans: GET requests retrieve data from the server and send parameters in the URL (visible). POST requests send data to the server in the request body (not visible in URL), typically used for form submissions or creating resources.

Q8. What are advantages of Node.js?

Ans: Key advantages: non-blocking asynchronous I/O (handles many requests without waiting), uses JavaScript on both front-end and back-end, large npm ecosystem, fast execution with V8 engine, good for real-time applications.

Q9. What does require() do in Node.js?

Ans: require() is used to import modules (built-in, npm packages, or custom files) into a Node.js file. For example: `const express = require('express')` imports the Express module so you can use it.

Q10. What is middleware in Express?

Ans: Middleware are functions that run during the request-response cycle in Express. They have access to req, res, and next. Examples: `express.static()` (serves files), `express.json()` (pares JSON body), custom authentication middleware.

Assignment 3b – CRUD API with Node.js, Express & MongoDB

Q1. What is MongoDB?

Ans: MongoDB is a NoSQL, document-oriented database that stores data as JSON-like BSON documents instead of rows and columns. It is schema-less, highly scalable, and ideal for web applications with flexible data structures.

Q2. What are CRUD operations?

Ans: CRUD stands for Create, Read, Update, Delete — the four basic database operations. In REST APIs: Create = POST, Read = GET, Update = PUT/PATCH, Delete = DELETE.

Q3. What is Mongoose?

Ans: Mongoose is an ODM (Object Data Modeling) library for MongoDB and Node.js. It provides a schema-based solution to model application data, with built-in type casting, validation, and query building.

Q4. What is a REST API?

Ans: A REST (Representational State Transfer) API is a set of HTTP endpoints that follow REST principles. Each endpoint represents a resource, and standard HTTP methods (GET, POST, PUT, DELETE) perform CRUD operations on it.

Q5. What is the difference between SQL and NoSQL databases?

Ans: SQL databases (like MySQL) use structured tables with fixed schemas and support JOIN operations. NoSQL databases (like MongoDB) store data as documents, key-value pairs, or graphs — offering flexibility, scalability, and schema-less design.

Q6. How does Express handle routing?

Ans: Express uses `app.get('/path', handler)`, `app.post('/path', handler)`, etc. to define routes. Route handlers receive request (`req`) and response (`res`) objects. Router middleware (`express.Router()`) organizes routes into separate files.

Q7. What is the purpose of `express.json()` middleware?

Ans: `express.json()` is middleware that parses incoming request bodies with JSON payloads. Without it, `req.body` would be undefined when the client sends JSON data in a POST or PUT request.

Q8. How do you connect Node.js to MongoDB?

Ans: Use Mongoose: `mongoose.connect('mongodb://localhost:27017/dbname')` in your Node.js file after installing mongoose via npm. The connection string specifies the host, port, and database name.

Assignment 4a – jQuery Mobile Website

Q1. What is jQuery Mobile?

Ans: jQuery Mobile is an HTML5-based user interface framework built on jQuery Core. It is designed to create responsive, touch-friendly websites and applications that work across mobile, tablet, and desktop devices with a consistent look and feel.

Q2. Why is jQuery Mobile used?

Ans: It allows developers to build web apps that work the same way on mobile, tablet, and desktop. It provides touch-friendly form inputs, UI widgets, and is compatible with frameworks like PhoneGap for native-like mobile apps.

Q3. What is the data-role attribute in jQuery Mobile?

Ans: data-role is an HTML attribute used by jQuery Mobile to identify UI components. Common values: data-role='page' (defines a page), data-role='header' (top bar), data-role='content' (main area), data-role='footer' (bottom bar), data-role='listview' (styled list).

Q4. What are the features of jQuery Mobile?

Ans: Features: Built on jQuery Core, open-source, cross-platform and cross-browser compatible, touch-optimized UI widgets, Ajax-powered page transitions, responsive design, progressive enhancement support.

Q5. How do you include jQuery Mobile in a project?

Ans: Include it via CDN in the HTML head: the jQuery library first, then the jQuery Mobile CSS link, then the jQuery Mobile JS script. Or download the ZIP from jquerymobile.com and include the CSS and JS files locally.

Q6. What is a single-page architecture in jQuery Mobile?

Ans: jQuery Mobile uses a single HTML file containing multiple 'pages' (divs with data-role='page'). Navigation between pages happens via Ajax transitions within the same file without full page reloads.

Assignment 4b – AWS Cloud Deployment

Q1. What is Cloud Computing?

Ans: Cloud computing means storing and accessing data and services over the internet instead of local hardware. It provides on-demand access to computing resources like servers, storage, databases, and software.

Q2. What is AWS?

Ans: AWS (Amazon Web Services) is Amazon's cloud computing platform. It offers 200+ services including compute (EC2), storage (S3), databases (RDS, DynamoDB), networking, and machine learning — with a pay-as-you-go pricing model.

Q3. What is Amazon EC2?

Ans: EC2 (Elastic Compute Cloud) is a service that provides resizable virtual servers (instances) in the cloud. You can launch Linux or Windows server instances to host web applications and configure CPU, RAM, and storage.

Q4. What is AWS Elastic Beanstalk?

Ans: Elastic Beanstalk is a PaaS (Platform as a Service) from AWS. You simply upload your application code, and Elastic Beanstalk automatically handles deployment, scaling, load balancing, and monitoring — without managing the infrastructure.

Q5. What is AWS VPC?

Ans: VPC (Virtual Private Cloud) is an isolated virtual network in AWS where you can launch resources in a logically separate section. You control IP address ranges, subnets, route tables, and network gateways for security and networking.

Q6. What are the AWS support plans?

Ans: AWS offers Basic Plan (free, self-service support), Developer Plan (paid, for testing/development), and Business Plan (paid, for production workloads with 24/7 support). Basic Plan is sufficient for learning purposes.

Q7. What is IaaS, PaaS, and SaaS?

Ans: IaaS (Infrastructure as a Service) provides virtualized hardware (e.g., EC2). PaaS (Platform as a Service) provides a managed platform for app deployment (e.g., Elastic Beanstalk). SaaS (Software as a Service) delivers complete applications over the internet (e.g., Gmail).

Q8. What are the deployment steps on AWS?

Ans: 1. Launch EC2 instance, 2. Configure source content/code, 3. Upload application files, 4. Deploy the application, 5. Test and update as needed, 6. Clean up resources to avoid charges.

Q9. When was AWS launched?

Ans: AWS services were first launched in 2002. Cloud products launched in 2006. By 2021, AWS comprised over 200 products and services and generates over \$10 billion in annual revenue.

General / Cross-Topic Oral Questions

Q1. What is a full stack web application?

Ans: A full stack web application involves both front-end (client-side: HTML, CSS, JavaScript, Angular) and back-end (server-side: Node.js, Express, APIs) development, along with a database (MongoDB). The full stack covers everything from UI to database.

Q2. What is the MVC architecture?

Ans: MVC (Model-View-Controller) separates an application into three parts: Model (data/business logic), View (user interface/presentation), Controller (handles user input and connects Model to View). It promotes clean, maintainable code.

Q3. What is the difference between front-end and back-end?

Ans: Front-end (client-side) is what users see and interact with — built with HTML, CSS, JavaScript, and frameworks like Angular/React. Back-end (server-side) processes requests, handles business logic, and manages databases — built with Node.js, Express, etc.

Q4. What is an API?

Ans: API (Application Programming Interface) is a set of rules that allows different software applications to communicate with each other. A Web API exposes endpoints (URLs) that clients call using HTTP to retrieve or manipulate data.

Q5. What is JSON?

Ans: JSON (JavaScript Object Notation) is a lightweight, human-readable data interchange format. It uses key-value pairs and arrays to represent data. Example: `{"name": "Alice", "age": 21}`. It is widely used in REST APIs.

Q6. What is the difference between HTTP GET and POST?

Ans: GET retrieves data from the server; parameters are passed in the URL, so they are visible and can be bookmarked. POST sends data to the server in the request body — more secure and suitable for form submissions and data creation.

Q7. What is a Single Page Application (SPA)?

Ans: A SPA loads a single HTML page and dynamically updates content as the user interacts with it — without full page reloads. Angular and React are popular SPA frameworks. They provide faster navigation and better user experience.

Q8. What is version control and why is it important?

Ans: Version control tracks all changes to code over time, allowing developers to revert mistakes, compare versions, and collaborate without overwriting each other's work. Git is the most widely used version control system.

Q9. What technologies are covered in this WAD subject?

Ans: WAD covers: HTML, CSS, Bootstrap (frontend), JavaScript and AJAX (scripting), Git/GitHub (version control), Docker (containerization), Angular (front-end framework), Node.js and Express (backend), MongoDB (database), jQuery Mobile (mobile web), and AWS (cloud deployment).

Q10. What is the difference between Angular and React?

Ans: Angular is a full-fledged MVC framework by Google using TypeScript with built-in tools for routing, forms, HTTP. React is a UI library by Facebook focused only on the View layer — more lightweight but requires additional libraries for routing and state management.